5.3.5 Authenticating AWS resources with roles

Various use cases exist where an EC2 instance needs to access or manage AWS resources. For example, an EC2 instance might need to do the following:

- Back up data to the object store S3
- Terminate itself after a job has been completed
- Change the configuration of the private network environment in the cloud

To be able to access the AWS API, an EC2 instance needs to authenticate itself. You could create an IAM user with access keys and store the access keys on an EC2 instance for authentication. But doing so is a hassle and violates security best practices, especially if you want to rotate the access keys regularly.

Instead of using an IAM user for authentication, you should use an IAM role whenever you need to authenticate AWS resources like EC2 instances. When using an IAM role, your access keys are injected into your EC2 instance automatically.

If an IAM role is attached to an EC2 instance, all identity policies attached to those roles are evaluated to determine whether the request is allowed. By default, no role is attached to an EC2 instance, and, therefore, the EC2 instance is not allowed to make any calls to the AWS API.

The following example will show you how to use an IAM role for an EC2 instance. Do you remember the temporary EC2 instances from